

Package #159

embed a passive, temperature■activated methane■generating chamber upstream of the check valve that releases gas on warm■up to create positive pressure and self■prime the syrup line without electronics

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'loss of prime' concerning syrup pump check valve. User-submitted engineering problem (session input): "Why do syrup-pump check valves in fountain soda machines lose prime and stall after warm restarts, and can a passively self-priming design fix it without electronics?". [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2024-2026). Illustrative records: 'Tracking water vapor homogeneous nucleation and droplet growth with spectroscopy and holography in a free expansion cloud chamber'; 'This report describes a series of tests performed at Sandia's burn site to better understand the behavior of aerosol commodities and their hazards in the transportation environment. This comes on the '. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: methane production by intestinal methanogen overgrowth (increase of methane■producing archaea)

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.